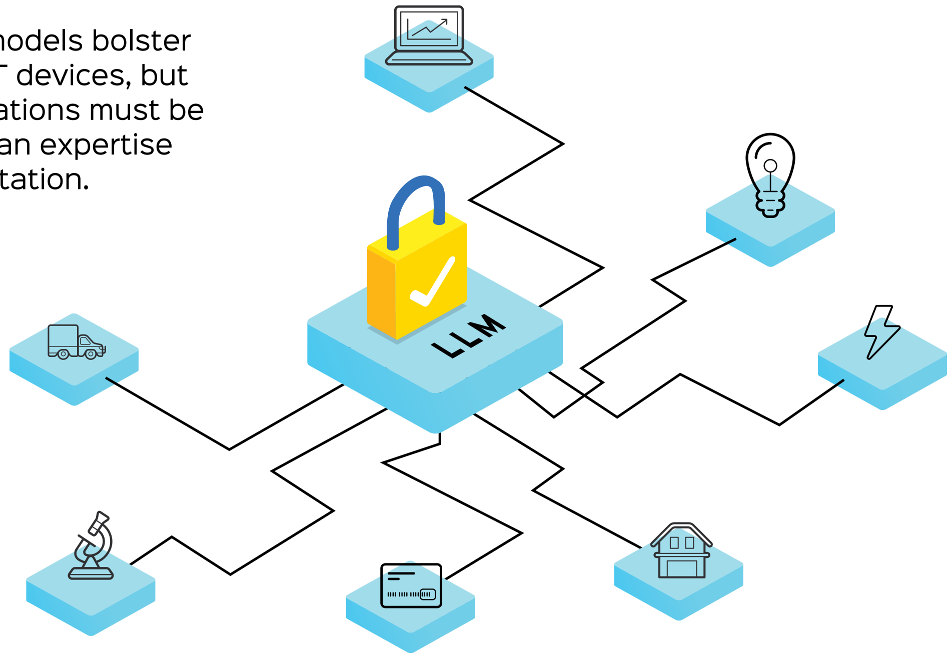
Large Language Models: Enhancing IoT Security

Large language models bolster the security of IoT devices, but their recommendations must be validated by human expertise prior to implementation.



The Internet of Things (IoT) is a network of physical objects that can connect and exchange data with other devices and systems over the internet. These can range from ordinary household objects to sophisticated industrial tools. IoT finds applications in various industries, including smart homes, wearable devices, smart cities, and Industrial IoT.

IoT is a rapidly growing field with the potential to revolutionise many industries and aspects of our lives. IoT devices can enhance our lives by increasing the efficiency and security of our homes and businesses, making our cities more livable, and facilitating proactive healthcare systems.

Why IoT security?

IoT devices play a critical role in essential infrastructure, such as energy grids, transportation systems, manufacturing plants and healthcare. Unauthorised access to these systems can result in severe repercussions, including power failures, transportation disruptions, and potential loss of life. Hence, protecting data, privacy, and the functionality of IoT devices is crucial.

In addition, IoT devices are often integrated into enterprise networks, creating a pathway for attackers to infiltrate and compromise corporate networks, leading to data breaches, and intellectual property theft.

Hence, as IoT devices proliferate, addressing IoT security challenges becomes increasingly important.

How is IoT security different from traditional security?

IoT security differs from traditional security in several ways due to the unique characteristics and challenges presented by IoT devices and ecosystems. Here are some of the primary differences between IoT security and traditional security.

- ***Attack surface:*** IoT devices are often connected to the internet, creating a much larger attack surface compared to traditional devices.
- ***Resource constraints:*** IoT devices frequently have limited computing, memory, network, and energy resources, which restrict the ability to implement complex security measures.
- ***Heterogeneity:*** IoT devices come from various manufacturers, featuring different operating systems and security features, making it challenging to uniformly implement traditional security measures.
- ***Physical environments:*** IoT devices are often deployed in diverse and difficult-to-access remote locations, making them susceptible to physical